

HW 2: Three-level systems



Question 1) Recall that a two-state system could be represented by a Bloch sphere ($\mathcal{H} = \mathbb{C}^2 \rightarrow \mathbb{CP}^1 \simeq S^2$), that is, a two-dimensional complex Hilbert space maps to a two-dimensional real space once we factor out the unphysical global phase. The two real parameters are usually represented by a polar and azimuthal angles, θ and ϕ , on the Bloch sphere. This describes pure states of spin-1/2 particles ($s = 1/2$, $m_s = \pm 1/2$) or qubits. For generic, mixed states (we will cover those later), the Bloch vector need not lie only on the spherical surface, but also in its interior.

We will now consider a three-state system with basis $\{|+\rangle, |0\rangle, |-\rangle\}$. You can imagine this as a spin-1 particle ($s = 1$, $m_s = -1, 0, +1$), or a “qutrit”. This system now requires a four real number parameterization, $\mathcal{H} = \mathbb{C}^3 \rightarrow \mathbb{CP}^2$. Unfortunately the latter cannot be represented by S^4 as you might be tempted to conclude. In fact, it turns out that to represent a generic (mixed) state in a three-state system, you’d need an 8-dimensional real space (as opposed to a 3-dimensional real space for the two-state system). This homework should convince you that while in 2-state systems we can reconstruct any state from its Bloch vector (three expectation values), in 3-state systems the Bloch vector is not enough, even for pure states. We will need 8 expectation values instead.

Any normalized physical state in a three-state system can be written as

$$|\psi\rangle = c_+ |+\rangle + c_0 |0\rangle + c_- |-\rangle, \quad (1)$$

with

$$|c_+|^2 + |c_0|^2 + |c_-|^2 = 1, \quad (2)$$

and

$$c_+ = \cos \frac{\theta}{2}, \quad c_0 = \sin \frac{\theta}{2} \sin \frac{\theta_0}{2} e^{i\phi_0}, \quad c_- = \sin \frac{\theta}{2} \cos \frac{\theta_0}{2} e^{i\phi}, \quad (3)$$

where $\theta_{(0)} \in [0, \pi]$ and $\phi_{(0)} \in [0, 2\pi)$.

- Show that fixing the polar angle $\theta_0 = 0$ leads to a subspace that recovers the two-state system (Bloch sphere) with $\{|+\rangle, |-\rangle\}$ as the basis kets. What are the polar and azimuthal angles that describe this subspace?
- What choice of angles gives a Bloch sphere with $\{|0\rangle, |-\rangle\}$ as the basis kets? How about $\{|0\rangle, |+\rangle\}$ as the basis kets? What are the polar and azimuthal angles that describe these subspaces? We have found three two-state subspaces within the three-state system, but there are infinitely many other possible two-state subspace combinations.
- Interpret this system as a spin-1 particle (here is where we start disconnecting from qutrits). We can define the x , y , and z components of the spin-1 operator as

$$\begin{aligned} \hat{S}_x &= \frac{\hbar}{\sqrt{2}} (|+\rangle \langle 0| + |0\rangle \langle +| + |0\rangle \langle -| + |-\rangle \langle 0|), \\ \hat{S}_y &= \frac{\hbar}{\sqrt{2}} (-i|+\rangle \langle 0| + i|0\rangle \langle +| - i|0\rangle \langle -| + i|-\rangle \langle 0|), \\ \hat{S}_z &= \hbar(|+\rangle \langle +| - |-\rangle \langle -|). \end{aligned} \quad (4)$$

Find the matrix form of these operators in the basis

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (5)$$

Express your answer as a 3×3 matrix as well as in terms of the Gell-Mann matrices, λ_i . These are the 3×3 analogs of the Pauli matrices, generators of the $\mathfrak{su}(3)$ algebra, the algebra of the $SU(3)$ group. Just like any operator could have been written in terms of the Pauli matrices for a two-state system, any operators in a three-state system can be written in terms of the Gell-Mann matrices.

- d) Check that the spin operators in Eq. (4) obey the same algebra as the spin- $\frac{1}{2}$ operators, $[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k$, and that $\hat{\mathbf{S}}^2 = s(s+1)\hbar^2\mathbb{1} = 2\hbar^2\mathbb{1}$. While generic three-state system operators are built on the $\mathfrak{su}(3)$ algebra, the spin operators themselves still obey the familiar $\mathfrak{su}(2)$ algebra.
- e) Calculate the Bloch vector

$$\mathbf{S} = \begin{pmatrix} \langle \hat{S}_x \rangle \\ \langle \hat{S}_y \rangle \\ \langle \hat{S}_z \rangle \end{pmatrix} \quad (6)$$

for $|+\rangle$, $|0\rangle$, $|-\rangle$, and the state

$$|s\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle). \quad (7)$$

These are all pure states. What can you say about the length of $\mathbf{S}$ for the $|0\rangle$ and $|s\rangle$ states? How do these two examples contrast with the Bloch vector for the two-state system?

- f) Calculate $\langle \hat{S}_{x,y,z}^2 \rangle$ and $\Delta S_{x,y,z} = \sqrt{\langle \hat{S}_{x,y,z}^2 \rangle - \langle \hat{S}_{x,y,z} \rangle^2}$ for $|+\rangle$, $|0\rangle$, $|-\rangle$, and $|s\rangle$. How does your result for $|0\rangle$ and $|s\rangle$ compare with the previous question? Can one infer the quantum state of the system solely from the Bloch vector for a three-state system? Convince yourself that this is the case for **two**-state systems.



Bonus) In the 2-state system, we needed 2 real numbers to specify a pure state. Any observable can be expressed in terms of the Pauli matrices, and knowing the Bloch vector allows us to determine the expectation value of any observable.

In three-state systems, we need only 4 real parameters to specify states. However, because observables can be expressed in terms of the Gell-Mann matrices, we would actually need to know 8 real parameters to fully determine the expectation values of all observables. The Bloch vector only gives us 3 out of 8. Here you will calculate the remaining 5.

The Bloch vector in three-state system is called the polarization vector. The rest lives in what is called the alignment tensor,

$$T_{ij} = \langle \hat{S}_i \hat{S}_j + \hat{S}_j \hat{S}_i \rangle - \frac{4}{3}\hbar^2\delta_{ij}, \quad i, j = x, y, z. \quad (8)$$

Note this is a symmetric, traceless rank-2 tensor.

- a) Prove this is a traceless tensor, that is,

$$\text{Tr}(T) = T_{xx} + T_{yy} + T_{zz} = 0. \quad (9)$$

This shows that we go from generic $3 \times 3 = 9$ degrees of freedom to 6 after imposing symmetry, and then 5 after imposing tracelessness. With the polarization vector, we have 3 more, recovering the total of 8.

- b) Calculate the components of the alignment tensor T_{ij} for the states $|0\rangle$ and $|s\rangle$.

HW 2 Solutions: Three-level systems

Start with the general state

$$|\psi\rangle = c_+ |+\rangle + c_0 |0\rangle + c_- |-\rangle, \quad c_+ = \cos \frac{\theta}{2}, \quad c_0 = \sin \frac{\theta}{2} \sin \frac{\theta_0}{2} e^{i\phi_0}, \quad c_- = \sin \frac{\theta}{2} \cos \frac{\theta_0}{2} e^{i\phi}. \quad (10)$$

You can check that it is normalized

$$|c_+|^2 + |c_0|^2 + |c_-|^2 = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} \left(\sin^2 \frac{\theta_0}{2} + \cos^2 \frac{\theta_0}{2} \right) = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1. \quad (11)$$

We have used the freedom of a common global phase which is why only two phases ϕ, ϕ_0 appear. The space of three-states is then described with four real parameters, as expected for $\mathbb{CP}^{3-1}$ (complex dimension 2 or real dimension 4).

a) Setting $\theta_0 = 0$ gives $c_0 = 0$ and

$$|\psi\rangle = \cos \left(\frac{\theta}{2} \right) |+\rangle + e^{i\phi} \sin \left(\frac{\theta}{2} \right) |-\rangle. \quad (12)$$

This is precisely the two-state parametrisation, with $|+\rangle$ at the north pole and $|-\rangle$ at the south pole. The polar angle is θ and the azimuthal angle is ϕ . The coordinate ϕ_0 is undefined on this subspace in the same sense that the azimuthal angle is undefined at the north and south poles.

b) For $\{|0\rangle, |-\rangle\}$, we need $c_+ = 0$ which is easily achieved with $\cos(\theta/2) = 0$, so that

$$|\psi\rangle = e^{i\phi_0} \sin \left(\frac{\theta_0}{2} \right) |0\rangle + e^{i\phi} \cos \left(\frac{\theta_0}{2} \right) |-\rangle = e^{i\phi} \left[\cos \left(\frac{\theta_0}{2} \right) |-\rangle + e^{i(\phi_0 - \phi)} \sin \left(\frac{\theta_0}{2} \right) |0\rangle \right], \quad (13)$$

where the phase $e^{i\phi}$ is unphysical. Again, this is a Bloch sphere with $|-\rangle$ at the north pole, polar angle θ_0 , and azimuthal angle $\phi_0 - \phi$.

Similarly for $\{|0\rangle, |+\rangle\}$, we require $c_- = 0$, achieved through $\theta_0 = \pi$:

$$|\psi\rangle = \cos \left(\frac{\theta}{2} \right) |+\rangle + e^{i\phi_0} \sin \left(\frac{\theta}{2} \right) |0\rangle, \quad (14)$$

with $|+\rangle$ at the north pole.

Note that there are infinitely many choices of linear combinations of $|+\rangle, |0\rangle, |-\rangle$ that can serve as the basis for a two-dimensional subspace, so there are infinitely many corresponding Bloch spheres inside the three-state system.

c) We will need to calculate the “matrix elements” in row m and column n : $\langle m | \hat{S}_i | n \rangle$. With each dyad $|m\rangle \langle n|$, we get a 1 in position (m, n) . Using the ordering $(+, 0, -)$ as suggested

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \hat{S}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (15)$$

You all found the Gell-Mann matrices,

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \quad (16)$$

By comparison,

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}} (\lambda_1 + \lambda_6), \quad \hat{S}_y = \frac{\hbar}{\sqrt{2}} (\lambda_2 + \lambda_7), \quad \hat{S}_z = \frac{\hbar}{2} (\lambda_3 + \sqrt{3} \lambda_8). \quad (17)$$

The $\sqrt{3}$ is ugly, but such is life in $\text{SU}(3)$.



There are eight Gell-Mann matrices, and the three spin operators use only three independent combinations of them. The five independent ones that are left over are exactly what you would need to build the alignment tensor in the bonus question.

d) There are two ways to answer this one: we can either use direct computation, as you all did, or you can use the Gell-Mann matrices. They respect their own $\mathfrak{su}(3)$ commutation relations:

$$[\lambda_a, \lambda_b] = 2i f_{abc} \lambda_c, \quad a, b, c = 1, \dots, 8. \quad (18)$$

The constants f_{abc} are called structure constants of the $\mathfrak{su}(3)$ Lie algebra. The analogous structure constants for the $\mathfrak{su}(2)$ Lie algebra

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k, \quad \text{for } \mathfrak{su}(2) \text{ algebra.} \quad (19)$$

are just the Levi-Civita symbols $f_{ijk} \rightarrow \epsilon_{ijk}$. Except for $\mathfrak{su}(3)$, the structure constants are more complicated:

$$f_{123} = 1, \quad f_{147} = -f_{156} = f_{246} = f_{257} = f_{345} = -f_{367} = \frac{1}{2}, \quad f_{458} = f_{678} = \frac{\sqrt{3}}{2}, \quad (20)$$

with all others being antisymmetric under the exchange of any two indices. If you want a challenge, you can try expanding the right hand side below to find the Levi-Civita tensor:

$$[\hat{S}_x, \hat{S}_y] = \frac{\hbar^2}{4} [\lambda_1 + \lambda_6, \lambda_2 + \lambda_7] = \frac{2i\hbar^2}{4} (f_{12c} \lambda_c + f_{16c} \lambda_c + f_{62c} \lambda_c + f_{67c} \lambda_c). \quad (21)$$

I will not reproduce the matrix multiplication here, but:

MATHEMATICA is your friend.

One set of matrices I will show are these:

$$\hat{S}_x^2 = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \hat{S}_y^2 = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad \hat{S}_z^2 = \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (22)$$

Note that these are not proportional to the identity matrix, unlike the spin- $\frac{1}{2}$ case. There we had $\hat{S}_i^2 = (\hbar/2\sigma_i)^2 = \frac{\hbar^2}{4} \mathbb{1}$, but now the spin-1 matrices are no longer Pauli operators and do not square to one. We find:

$$\hat{\mathbf{S}}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = 2\hbar^2 \mathbb{1} = s(s+1)\hbar^2 \mathbb{1} \quad \text{with } s = 1. \quad (23)$$

e) Now to calculate the Bloch vector. It will be instructive to do so in all generality, using the general form of the bloch vector in Eq. (1). Carrying this out, we find

$$\langle \hat{S}_x \rangle = \sqrt{2} \hbar \text{Re}[c_0^* (c_+ + c_-)], \quad \langle \hat{S}_y \rangle = -\sqrt{2} \hbar \text{Im}[c_0^* (c_+ - c_-)], \quad \langle \hat{S}_z \rangle = \hbar (|c_+|^2 - |c_-|^2). \quad (24)$$

I do not write out the angles because it is very messy and does not really help us. Evaluating for the four states requested:

$$\langle \hat{\mathbf{S}} \rangle_+ = \langle + | \hat{\mathbf{S}} | + \rangle = \begin{pmatrix} \langle \hat{S}_x \rangle_+ \\ \langle \hat{S}_y \rangle_+ \\ \langle \hat{S}_z \rangle_+ \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \hbar \end{pmatrix}. \quad (25)$$

Same for $|0\rangle$, $|-\rangle$, and $|s\rangle$:

$$\langle \hat{\mathbf{S}} \rangle_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \langle \hat{\mathbf{S}} \rangle_- = \begin{pmatrix} 0 \\ 0 \\ -\hbar \end{pmatrix}, \quad \langle \hat{\mathbf{S}} \rangle_s = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (26)$$

Two key points: i) the length of the Bloch vector is not fixed, and ii) the states $|0\rangle$ and $|s\rangle$ have the same Bloch vector:

$$\langle \hat{\mathbf{S}} \rangle_s = \langle \hat{\mathbf{S}} \rangle_0. \quad (27)$$

Contrast this with the two-state system, where we proved that (see two-state system notes):

$$\sum_i \langle \hat{S}_i \rangle^2 = \frac{\hbar^2}{4} \quad (28)$$

for every pure state.

Another conclusion: a pure two-state Bloch vector always points somewhere and has a fixed length $\hbar/2$. A three-state “Bloch vector” need not point anywhere at all.

i

The general statement, valid for any spin, is

$$\sum_i (\Delta S_i)^2 = \langle \hat{\mathbf{S}}^2 \rangle - |\mathbf{S}|^2 = s(s+1)\hbar^2 - |\mathbf{S}|^2. \quad (29)$$

For $s = \frac{1}{2}$ the last term is always $\hbar^2/4$, but not $s = 1$.

f) From Eq. (22),

$$\langle \hat{S}_{x,y}^2 \rangle = \frac{\hbar^2}{2} (1 + |c_0|^2 \pm 2 \operatorname{Re}[c_+^* c_-]), \quad \langle \hat{S}_z^2 \rangle = \hbar^2 (1 - |c_0|^2), \quad (30)$$

with the upper sign for x . Using this, you can go through the calculation of the variance $\Delta S_i^2 = \langle \hat{S}_i^2 \rangle - \langle \hat{S}_i \rangle^2$ for each component. Putting it all into a “variance vector”, we have

$$\langle \Delta \mathbf{S}^2 \rangle_\psi = \begin{pmatrix} \langle \Delta S_x^2 \rangle_\psi \\ \langle \Delta S_y^2 \rangle_\psi \\ \langle \Delta S_z^2 \rangle_\psi \end{pmatrix}, \quad (31)$$

such that

$$\langle \Delta \mathbf{S}^2 \rangle_0 = \begin{pmatrix} \hbar^2/2 \\ \hbar^2/2 \\ \hbar^2 \end{pmatrix}, \quad \langle \Delta \mathbf{S}^2 \rangle_+ = \begin{pmatrix} \hbar^2/2 \\ \hbar^2/2 \\ 0 \end{pmatrix}, \quad \langle \Delta \mathbf{S}^2 \rangle_- = \begin{pmatrix} \hbar^2/2 \\ \hbar^2/2 \\ \hbar^2 \end{pmatrix}, \quad \langle \Delta \mathbf{S}^2 \rangle_s = \begin{pmatrix} 0 \\ \hbar^2 \\ \hbar^2 \end{pmatrix}, \quad (32)$$

We can also calculate the norm of this variance vector

$$\sum_i \langle \Delta S_i^2 \rangle_\psi = \langle \Delta S_x^2 \rangle_\psi + \langle \Delta S_y^2 \rangle_\psi + \langle \Delta S_z^2 \rangle_\psi \quad (33)$$

to get

$$\sum_i \langle \Delta S_i^2 \rangle_0 = 2\hbar^2, \quad \sum_i \langle \Delta S_i^2 \rangle_+ = \hbar^2, \quad \sum_i \langle \Delta S_i^2 \rangle_- = 2\hbar^2, \quad \sum_i \langle \Delta S_i^2 \rangle_s = 2\hbar^2. \quad (34)$$

Note in particular that

$$\langle \Delta \mathbf{S}^2 \rangle_0 \neq \langle \Delta \mathbf{S}^2 \rangle_s. \quad (35)$$

Clearly, even though these two states share the same Bloch vector, their variances are different; their uncertainties are distributed differently among the spin components. So, it must be that the Bloch vector does **not** uniquely determine the state.

i

Let's convince ourselves that for a qubit, it can. Any pure qubit state may be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |+\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |-\rangle, \quad (36)$$

for which

$$\mathbf{S} = \frac{\hbar}{2} (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta). \quad (37)$$

Given $\mathbf{S}$ we read off $\cos\theta = 2\langle \hat{S}_z \rangle / \hbar$ and $\tan\varphi = \langle \hat{S}_y \rangle / \langle \hat{S}_x \rangle$, which fix θ and φ uniquely and hence fix $|\psi\rangle$ up to the unphysical global phase.

Bonus a) Set $i = j$ in

$$T_{ij} = \langle \hat{S}_i \hat{S}_j + \hat{S}_j \hat{S}_i \rangle - \frac{4}{3} \hbar^2 \delta_{ij} \quad (38)$$

and sum over i . The first term gives $\sum_i \langle 2\hat{S}_i^2 \rangle = 2\langle \hat{\mathbf{S}}^2 \rangle = 4\hbar^2$ by Eq. (23), and the second gives $\frac{4}{3} \hbar^2 \delta_{ii} = \frac{4}{3} \hbar^2 \cdot 3 = 4\hbar^2$. Hence

$$\text{Tr } T = T_{xx} + T_{yy} + T_{zz} = 4\hbar^2 - 4\hbar^2 = 0. \quad (39)$$

Bonus b) One can check that all off-diagonal components vanish. The diagonal ones follow immediately from our results in (f), so that $T_{ii} = 2\langle \hat{S}_i^2 \rangle - \frac{4}{3} \hbar^2$ (no sum over i). Let me clarify the notation a bit: we can define the operator

$$\hat{T}_{ij} = \hat{S}_i \hat{S}_j + \hat{S}_j \hat{S}_i - \frac{4}{3} \hbar^2 \delta_{ij} \hat{\mathbb{1}}, \quad (40)$$

such that the tensor we care about is simply the expectation value of this operator:

$$T_{ij} = \langle \hat{T}_{ij} \rangle. \quad (41)$$

In this way,

$$\langle T \rangle_0 = \hbar^2 \begin{pmatrix} \frac{2}{3} & 0 & 0 \\ 0 & \frac{2}{3} & 0 \\ 0 & 0 & -\frac{4}{3} \end{pmatrix}, \quad \langle T \rangle_s = \hbar^2 \begin{pmatrix} -\frac{4}{3} & 0 & 0 \\ 0 & \frac{2}{3} & 0 \\ 0 & 0 & \frac{2}{3} \end{pmatrix}. \quad (42)$$

Both are traceless, as they must be, but not quite the same. The state $|0\rangle$ is said to be “aligned” along the z -axis, while the state $|s\rangle$ is aligned along the x -axis.